What are key tasks for data mining?

Abnormaly detection：排除異常值

Association rule learning：使用獨立性陽強的欄位，例如：帥、鼻子挺、眼睛大 都在描述同一件事

Clustering：分群（非監督學習??）

Classification：分類（監督學習??）

Regression analysis：

Summarization：

五大名詞

Accuracy:

Precision

Recall

Sensitivity

Specificity

Classify：分類

Regression：給數值

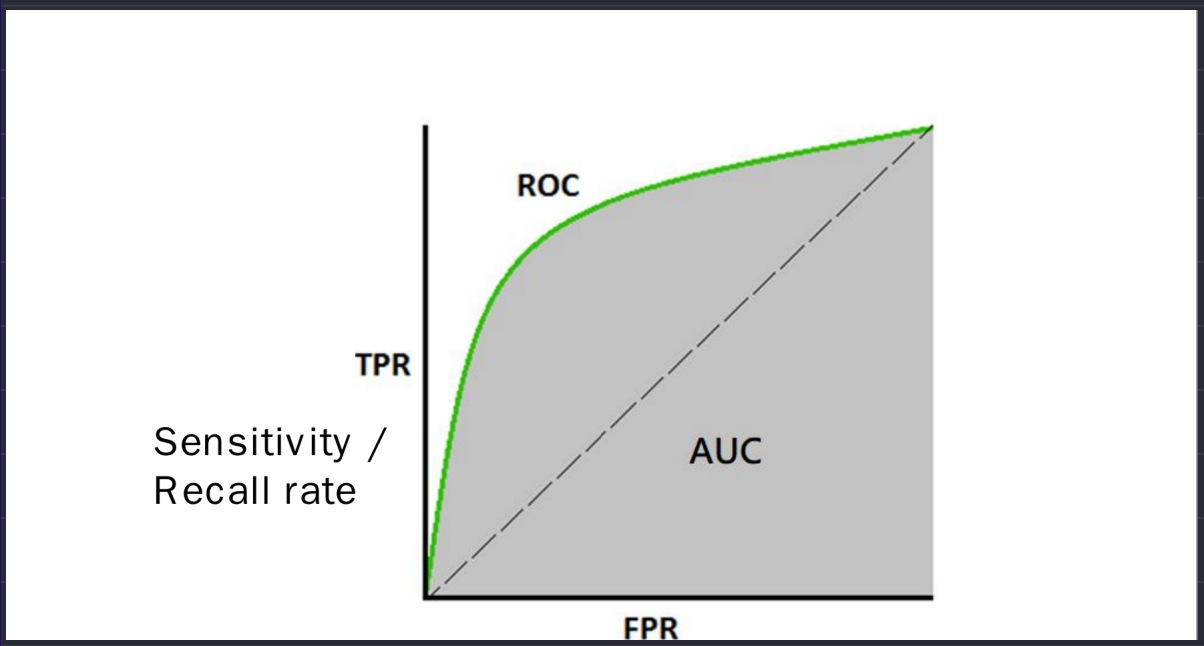
Evaluating Classification (預測分類) Results

Confusion matrix (a.k.a error matrix)

		True condition				
	Total population	Condition positive	Condition negative	Prevalence $= \frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) $= \frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	**True positive**, Power	**False positive**, Type I error	Positive predictive value (PPV), Precision $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR−}}$	F_1 score $= \frac{1}{\frac{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}{2}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) $= \frac{\text{FNR}}{\text{TNR}}$		

小考：數值有幾種分類？ 五種

TODO:



Evaluating Regression (預測數值) Results

Mean Squared Error (MSE)

$$MSE = \frac{1}{N}\sum_{j=1}^{N}(y_j - \check{y}_j)^2$$

Mean Absolute Error (MAE)

$$MAE = \frac{1}{N}\sum_{j=1}^{N}|y_j - \check{y}_j|$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(y_j - \check{y}_j)^2}$$

R² (R-Squared)

$$SE(\bar{Y}) = \sum_{i=1}^{N}(y_i - \bar{y})^2 \quad SE(line) = \sum_{j=1}^{N}(y_j - \check{y}_j)^2 \quad coeff(R^2) = 1 - \frac{SE(line)}{SE(\bar{Y})}$$

Data mining Techniques

Types of Data Mining / Machine Learning

1. Supervised Machine Learning
   (a) Classification：分類、判斷是非
   (b) Regression：預測、推斷數值
2. Unsupervised Machine Learning
   (a) Clustering：分群
   (b) Association：相關 ?_?
   (c) Dimensionality Reduction ：維度降低、加速效率
3. Semi-supervised learning (半監督式學習)
4. Reinforcement Learning